Udemy ML, DL, NLP Course (1)
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sec ~~Dec~~ -44 Anomaly
Detection Machine
learning

lec 1 Anomaly Detection
by Isolation Forest

lec 2: Local outlier
factor Anomaly
Detection

Anomaly Detection [to detect
outlier]
↳ plays an
important
role



Anomaly are basically the detection of outliers which is caused leading to the disruption in data.

Q. For example, we take the example of ISL.

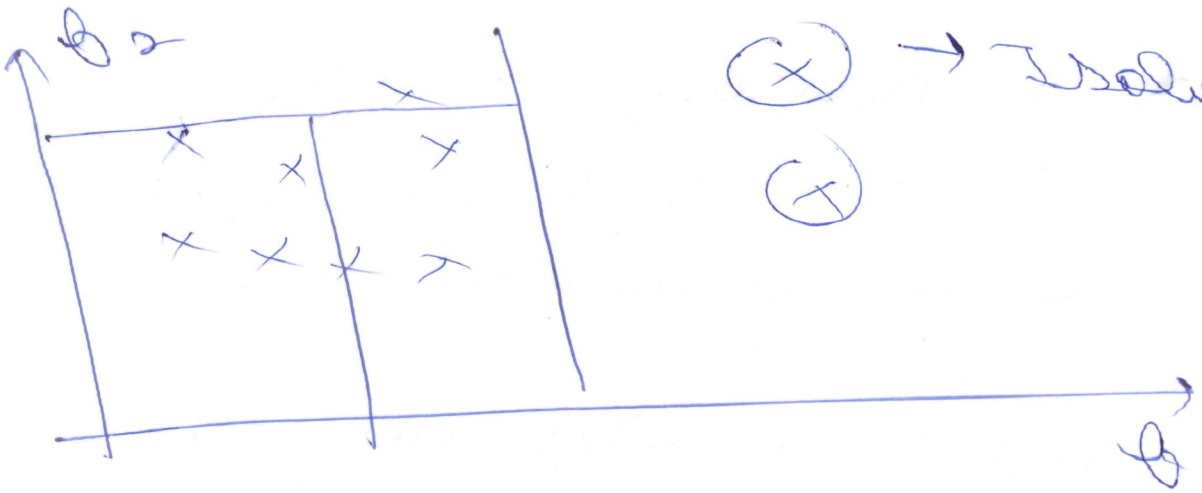
ISL

1	15
2	10
3	12
4	100

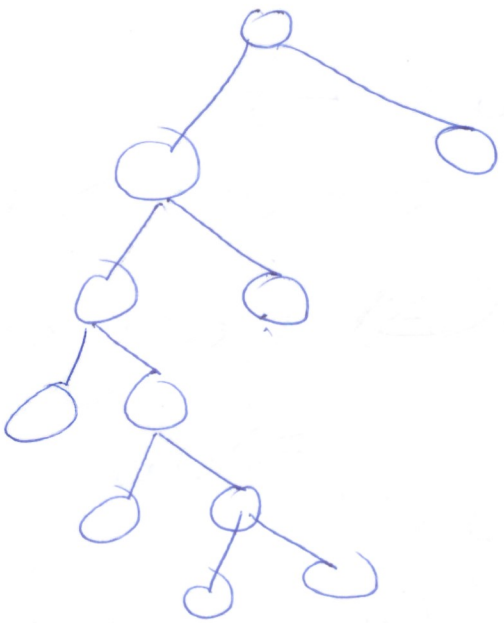
This is an anomaly as there can't be > 36

⇒ Isolation Forest (Decision Forest)

b1 b2 b3 b4



→ Isolated



→ Isolated as a leaf node

Anomaly score : If the Anomaly score crosses the threshold, it becomes an outlier

$$S(x, n) = 2^{\frac{-E(h(x))}{c(n)}}$$

n — No. of data in the

x — Outer nodes

$E(h(x))$ = Average search depth from the isolated tree

$c(n)$ = Average depth of $h(x)$

$E(h(x)) << c(n) \Rightarrow$

$s(x, n) \approx 1$

[threshold ≥ 0.5] $\Rightarrow$ Anomaly score $\Rightarrow$ outliers

$E(h(x)) << c(n)$

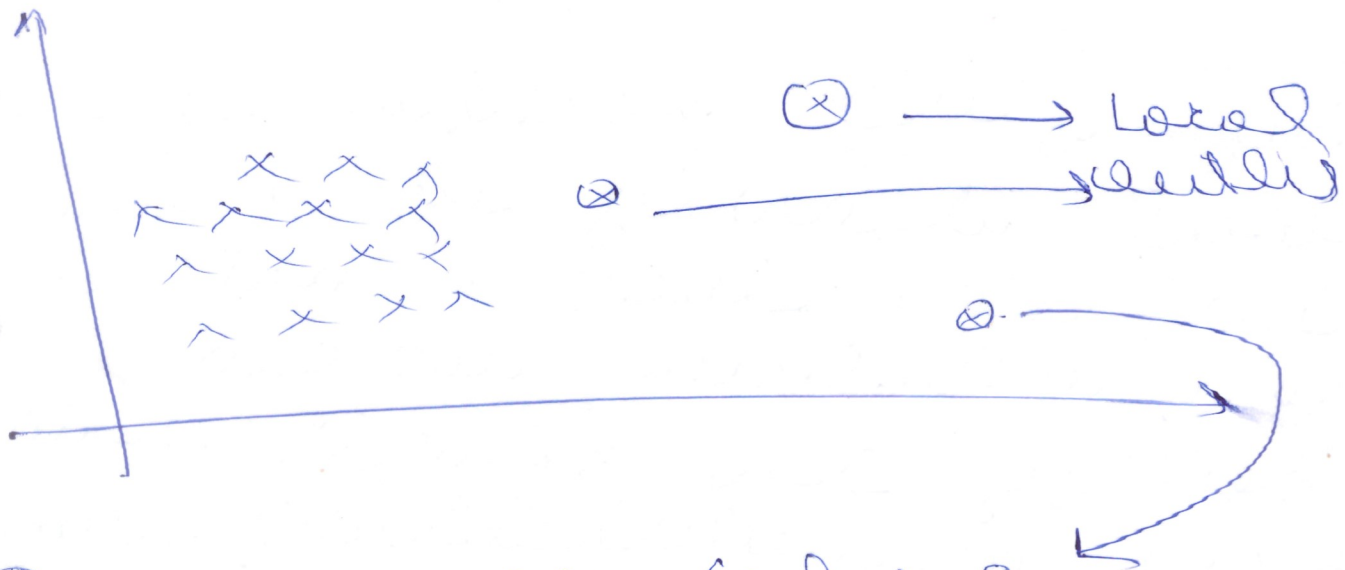
$\Rightarrow s(x, n) \approx 1$

$\Rightarrow$ Anomaly score $\Rightarrow$ outliers

$E(h(x)) >> c(n) \Rightarrow s(x, n) \approx 0$

$\Rightarrow$ Normal data points

Local outlier factor namely detection



1. Local outlier & Global outlier
2. Global outlier

Global outlier is completely different

Local outlier
Global outlier

[LOF] → Log score

It uses K Nearest Neighbour

↓

Local density

suppose $K = 5$, then we take a point is around the 5 neighbouring points from it.

- By just comparing the local density of a sample to the local densities of neighbours, we can identify the samples that have a substantially lower density than neighbours